

SAT Math Mastery · Week 3A

Functions & Graph Reading

Time	150 minutes: 110 min solving + 40 min self-check and error log
Topics	function notation · composition · transformations of graphs and tables · reading graphs, crossings and zeros · $f(x) = k$ solution counts · domain and range · piecewise and graph-given functions · average rate of change
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Evaluate $f(a)$, $f(a + h)$, $f(kx)$, $f(g(x))$ from a formula, a table, or a graph.
- S2.** Read a graph: values $f(a)$, intercepts, maxima/minima, and the inputs that give a given output.
- S3.** Count solutions of $f(x) = k$ (and $f(x) = g(x)$) by sliding a horizontal line across the graph.
- S4.** Compose functions in either order and spot when $f(g(x)) \neq g(f(x))$.
- S5.** Transform graphs and tables: $f(x) + k$, $f(x - h)$, $af(x)$, $f(kx)$, and recover the equation of a transformed graph.
- S6.** Use symmetry and equal outputs: $f(a) = f(b)$ and $f(x) = f(c - x)$.
- S7.** Find domain restrictions (zero denominators, even roots) and the effect of a stated input interval.
- S8.** Compute average rate of change $\frac{f(b) - f(a)}{b - a}$ from a graph, table, or formula.

2 · REQUIRED FACTS

- $f(a)$ means: input a , output $f(a)$. On a graph it is the *height* of the curve above $x = a$.
- A graph passes the vertical-line test; the equation $f(x) = k$ is solved by drawing the horizontal line $y = k$ and counting intersections.
- $f(x) = g(x)$ has as many solutions as the graphs of f and g have common points; touching a curve counts once.
- Composition works inside out: $f(g(x))$ means evaluate g first, then feed that output into f .
- Replacing x by $x - h$ moves the graph h units *right*; adding k outside moves it k units *up*.
- $f(x) = f(c - x)$ for all x means the graph is symmetric about the vertical line $x = \frac{c}{2}$.
- A denominator can never be 0 and an even root needs a non-negative radicand; a stated interval can remove more inputs.
- Closed dot: endpoint included. Open dot: endpoint excluded.

3 · REQUIRED FORMULAS

$$(f \circ g)(x) = f(g(x)), \quad g(x) = f(x - h) + k \Rightarrow \text{every point } (a, b) \text{ moves to } (a + h, b + k)$$

$$\text{average rate of change on } [a, b] = \frac{f(b) - f(a)}{b - a}, \quad f(x) = f(c - x) \Rightarrow \text{axis } x = \frac{c}{2}$$

$$\text{even/odd: } f(-x) = f(x) \text{ (mirror in the } y\text{-axis)}, \quad af(x) : \text{stretches } y\text{-values by } a \text{ (reflects if } a < 0)$$

4 · MINI THEORY SHEET

Reading a function from each representation.

Given	To find $f(a)$ / solve $f(x) = k$
formula	substitute a (use parentheses around negatives); solve the equation for x
table	find the row with input a ; look for rows with output k (there may be several)
graph	go to $x = a$ and read the height; draw $y = k$ and read every crossing's x
piecewise	first decide which branch the input belongs to, then substitute

Composition from a graph or table. Do it in two reads: the inner output becomes the next input. Write the intermediate value down.

Transformations. Inside the bracket the change is horizontal and *opposite*: $f(x + 3)$ moves left 3. Outside the bracket it is vertical and *as written*: $f(x) - 2$ moves down 2. Do the inside change first when both appear.

Equal outputs. If $f(a) = f(b)$ for a function with a symmetry axis, then the axis is $x = \frac{a+b}{2}$.

Counting crossings. Count each distinct x ; a tangent contact is one point, a vertex that sits exactly on the line is one point, endpoints matter only when they are closed dots.

Function notation in context. $f(5) = 105$ is one input–output pair. $f(t + 3) - f(t)$ is the change over 3 units of input. $f(t + 1)/f(t)$ is the growth factor for exponentials.

Speed shortcuts.

Situation	Shortcut
$f(g(x))$ with numbers	work inside out and write the middle value
$f(ax + b) = \dots$ and need $f(c)$	set $ax + b = c$, solve, substitute (do not find $f(x)$ first)
$f(a) = f(b)$ (parabola, abs. value)	axis (or vertex) at $x = \frac{a+b}{2}$
average rate of change of $px^2 + qx + r$ on $[a, b]$	$p(a + b) + q$ (no need to evaluate)
$f(x) = k$ count	horizontal line $y = k$ across the graph; ignore parts outside the domain
$f(f(x)) = c$ count	solve $f(u) = c$ first, then count $f(x) = u$ for every valid u
$f(t + d)/f(t)$ for ab^t	always b^d , whatever t is
transformed graph $\rightarrow$ equation	track the key point (vertex or intercept), then match a from one other point
undefined inputs	set each denominator factor to 0

5 · GUIDED EXAMPLES

Example 1 — Composition and function notation

$f(x) = x^2 - 3x$ and $g(x) = 2x + 1$. Find $f(g(2))$ and $g(f(2))$.

$g(2) = 5$, then $f(5) = 25 - 15 = 10$. So $f(g(2)) = 10$.

$f(2) = 4 - 6 = -2$, then $g(-2) = -4 + 1 = -3$. So $g(f(2)) = -3$. Order matters.

Example 2 — Counting solutions of $f(x) = k$

The graph of $y = f(x)$ is a parabola with vertex $(1, 6)$ opening downward, on the domain $-2 \leq x \leq 4$ (closed dots), with $f(-2) = -3$ and $f(4) = -3$. How many values of x satisfy $f(x) = 0$?

The height 0 lies below the vertex (6) and above the endpoint heights (-3), so the line $y = 0$ cuts the arch twice: **2 solutions**. Had the line been $y = 6$ it would touch once; $y = -5$ misses the graph entirely.

Example 3 — Transformations and average rate of change

The point $(3, -2)$ is on $y = f(x)$. The function g is $g(x) = 2f(x - 1) + 5$. Which point is on $y = g(x)$, and what is the average rate of change of $h(x) = x^2 + 4x$ on $[1, 5]$?

$x \rightarrow x + 1$ (shift right), $y \rightarrow 2y + 5$: $(3, -2) \rightarrow (4, 2(-2) + 5) = (4, 1)$.

Shortcut: $p(a + b) + q = 1(1 + 5) + 4 = 10$. Check: $\frac{h(5) - h(1)}{4} = \frac{45 - 5}{4} = 10$.

6 · PRACTICE SET — 34 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 3x^2 - 2x$. What is the value of $f(-2)$?

(A) 8

(B) 16 ✓

(C) -16

(D) 40

1. $f(-2) = 3(-2)^2 - 2(-2)$.

2. $3(4) = 12$ and $-2(-2) = +4$.

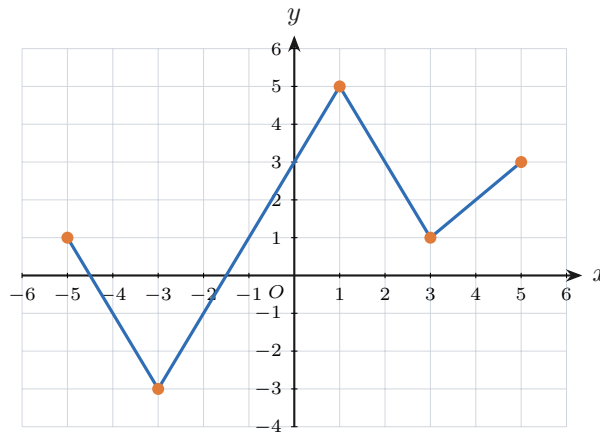
3. $12 + 4 = 16$.

Answer: (B)

2.

EASY · NO CALCULATOR NEEDED

The graph of $y = f(x)$ is shown in the xy -plane. What is the value of $f(0) + f(3)$?



1. At $x = 0$ the graph crosses the y -axis at height 3, so $f(0) = 3$.

2. At $x = 3$ the graph is at the point $(3, 1)$, so $f(3) = 1$.

3. $f(0) + f(3) = 3 + 1 = 4$.

Answer: 4

3.

EASY · NO CALCULATOR NEEDED

The table gives values of the function f . What is the value of $f(f(2))$?

x	1	2	3	4
$f(x)$	3	4	1	2

(A) 4

(B) 1

(C) 2 ✓

(D) 3

1. Inner value: $f(2) = 4$.2. Outer value: $f(4) = 2$.3. So $f(f(2)) = 2$.**Answer: (C)**

4.

EASY · NO CALCULATOR NEEDED

The function f is defined by $f(x) = x^2 - 4x$. The graph of $y = g(x)$ is the graph of $y = f(x)$ translated 3 units up. Which of the following defines g ?

(A) $g(x) = x^2 - 4x + 3$ ✓(B) $g(x) = x^2 - 4x - 3$ (C) $g(x) = (x + 3)^2 - 4(x + 3)$ (D) $g(x) = (x - 3)^2 - 4(x - 3)$ 1. Translating up 3 adds 3 to every output: $g(x) = f(x) + 3$.2. $g(x) = x^2 - 4x + 3$.3. Replacing x by $x \pm 3$ would be a horizontal shift, so the last two options are not vertical translations.**Answer: (A)**

5.

EASY · NO CALCULATOR NEEDED

The function g is defined by $g(x) = \frac{12}{x-3}$. For which of the following values of x is $g(x)$ undefined?

(A) 12

(B) -3

(C) 0

(D) 3 ✓

1. A fraction is undefined when its denominator is 0.

2. $x - 3 = 0 \Rightarrow x = 3$.**Answer: (D)**

6.

MEDIUM · NO CALCULATOR NEEDED

Functions f and g are defined by $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. What is the value of $f(g(2)) - g(f(2))$?

1. $g(2) = 2^2 + 1 = 5$, so $f(g(2)) = f(5) = 3(5) - 2 = 13$.

2. $f(2) = 3(2) - 2 = 4$, so $g(f(2)) = g(4) = 4^2 + 1 = 17$.

3. $13 - 17 = -4$.

Answer: -4

7.

MEDIUM · NO CALCULATOR NEEDED

The function f is defined by $f(x) = (x - 2)(x + 6)$. If $f(m) = f(-3)$ and $m \neq -3$, what is the value of m ?

1. The zeros are 2 and -6 , so the axis of symmetry is $x = \frac{2 + (-6)}{2} = -2$.

2. Equal outputs come from inputs equally far from the axis. -3 is 1 unit left of -2 , so m is 1 unit right: $m = -1$.

3. Check: $f(-3) = (-5)(3) = -15$ and $f(-1) = (-3)(5) = -15$.

Answer: -1

8.

MEDIUM · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 3x^2 - x$. Which of the following expressions is equivalent to $f(2x)$?

(A) $12x^2 - x$

(B) $6x^2 - x$

(C) $12x^2 - 2x$ ✓

(D) $6x^2 - 2x$

1. Replace every x with $2x$: $f(2x) = 3(2x)^2 - (2x)$.

2. $(2x)^2 = 4x^2$, so $3(4x^2) = 12x^2$; and $-(2x) = -2x$.

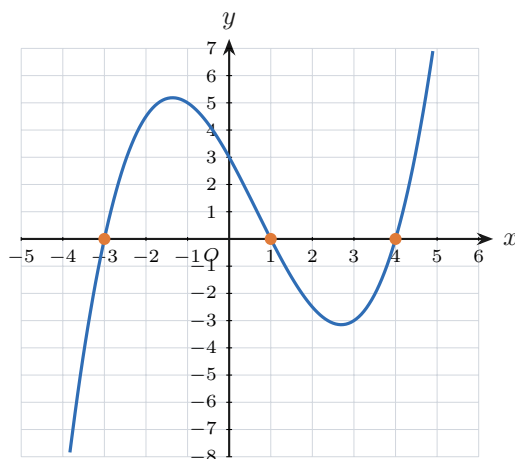
3. $f(2x) = 12x^2 - 2x$.

Answer: (C)

9.

MEDIUM · NO CALCULATOR NEEDED

The graph of $y = f(x)$ is shown in the xy -plane. For how many values of x does $f(x) = 2$?



(A) 2

(B) 3 ✓

(C) 4

(D) 1

1. Draw the horizontal line $y = 2$ across the graph.
2. On the left the curve climbs through height 2 (once), after its peak it falls through height 2 (second time), and after the valley it rises through height 2 again (third time).
3. The line meets the graph 3 times, so there are 3 values of x .

Answer: (B)

10.

MEDIUM · NO CALCULATOR NEEDED

The table shows values of the function f . The function g is defined by $g(x) = f(x - 2) + 1$. What is the value of $g(5)$?

x	1	3	5	7
$f(x)$	4	-2	6	9

(A) 10

(B) -3

(C) 7

(D) -1 ✓

1. $g(5) = f(5 - 2) + 1 = f(3) + 1$.
2. From the table $f(3) = -2$.
3. $g(5) = -2 + 1 = -1$.

Answer: (D)

11.

MEDIUM · CALCULATOR

The function $p(n) = 18n - 240$ models the profit, in dollars, from selling n tickets to a school fundraiser. What is the least whole number of tickets that must be sold so that the profit is positive?

1. Need $p(n) > 0$: $18n - 240 > 0 \Rightarrow n > \frac{240}{18} \approx 13.33$.
2. n must be a whole number, so the least value is 14.
3. Check: $p(13) = -6 < 0$ and $p(14) = 12 > 0$.

Answer: 14

12.

MEDIUM · NO CALCULATOR NEEDED

The function f is defined by $f(x) = \begin{cases} 2x + 1, & x < 3 \\ x^2 - 4, & x \geq 3 \end{cases}$. What is the value of $f(f(1))$?

(A) 5 ✓

(B) 7

(C) 3

(D) -5

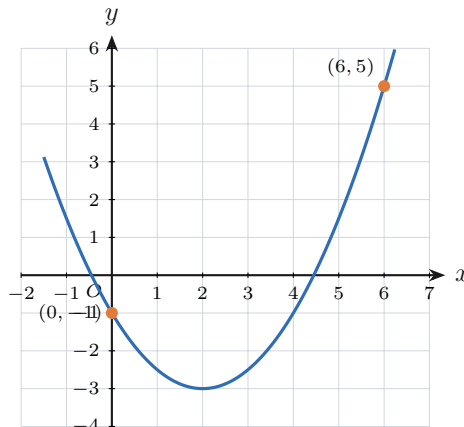
1. Since $1 < 3$ use $2x + 1$: $f(1) = 3$.
2. Now $f(3)$: $3 \geq 3$, so use $x^2 - 4$: $f(3) = 9 - 4 = 5$.
3. $f(f(1)) = 5$.

Answer: (A)

13.

MEDIUM · NO CALCULATOR NEEDED

The graph of $y = f(x)$ is shown, with two points labeled. What is the average rate of change of f from $x = 0$ to $x = 6$?



(A) 5

(B) 1 ✓

(C) 2

(D) 6

1. Read $f(0) = -1$ and $f(6) = 5$.

2.
$$\frac{f(6) - f(0)}{6 - 0} = \frac{5 - (-1)}{6} = \frac{6}{6} = 1.$$

Answer: (B)

14.

HARD · NO CALCULATOR NEEDED

Functions f and g are defined by $f(x) = 2x - 1$ and $g(x) = x^2 + 2$. There are two values of a for which $f(g(a)) = g(f(a))$. What is the positive one?

1. $f(g(a)) = 2(a^2 + 2) - 1 = 2a^2 + 3.$

2. $g(f(a)) = (2a - 1)^2 + 2 = 4a^2 - 4a + 3.$

3. Set equal: $2a^2 + 3 = 4a^2 - 4a + 3 \Rightarrow 0 = 2a^2 - 4a = 2a(a - 2).$

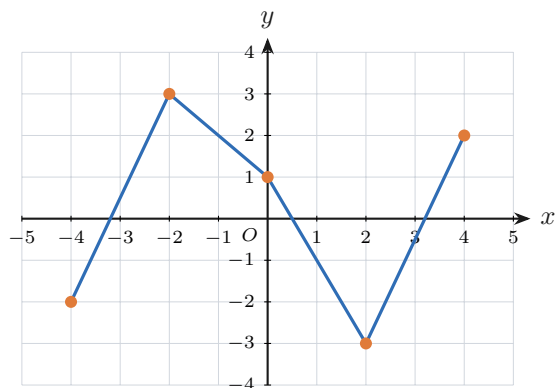
4. $a = 0$ or $a = 2$; the positive value is 2.

Answer: 2

15.

HARD · NO CALCULATOR NEEDED

The graph of the piecewise linear function f is shown, with the vertices at grid points. What is the value of $f(f(0)) + f(f(1))$?



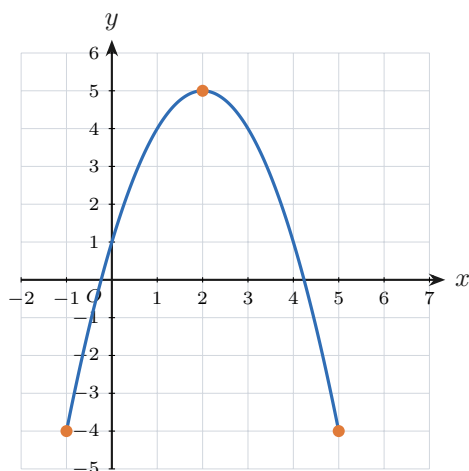
1. $f(0) = 1$ (the vertex $(0, 1)$). On the segment from $(0, 1)$ to $(2, -3)$ the slope is -2 , so $f(1) = -1$.
2. $f(f(0)) = f(1) = -1$.
3. $f(f(1)) = f(-1)$. On the segment from $(-2, 3)$ to $(0, 1)$ the slope is -1 , so $f(-1) = 2$.
4. Sum: $-1 + 2 = 1$.

Answer: 1

16.

HARD · DESMOS

The graph of $y = f(x)$ is shown. It is a portion of a parabola on the interval $-1 \leq x \leq 5$, and the endpoints are closed dots. For how many integer values of k does the equation $f(x) = k$ have exactly two solutions?



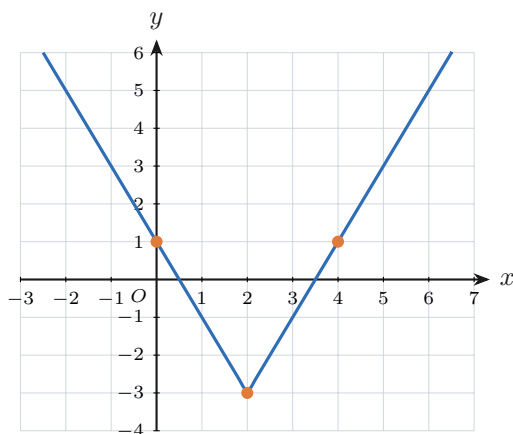
1. The vertex is $(2, 5)$ and both endpoints are at height -4 : $f(-1) = f(5) = -4$.
2. For $-4 < k < 5$ the line $y = k$ cuts the arch twice. For $k = 5$ it touches once. For $k = -4$ it meets the two (closed) endpoints, so it still gives two solutions. For $k < -4$ it misses the graph.
3. Integers with exactly two solutions: $k = -4, -3, \dots, 4$, which is 9 values.

Answer: 9

17.

HARD · NO CALCULATOR NEEDED

The graph of $y = g(x)$ is shown. Which of the following defines g ?



(A) $g(x) = 4|x - 2| - 3$

(B) $g(x) = 2|x - 2| + 3$

(C) $g(x) = 2|x - 2| - 3$ ✓

(D) $g(x) = 2|x + 2| - 3$

1. The vertex is $(2, -3)$, so $g(x) = a|x - 2| - 3$.
2. The point $(4, 1)$ is on the graph: $1 = a|4 - 2| - 3 \Rightarrow 2a = 4 \Rightarrow a = 2$.
3. $g(x) = 2|x - 2| - 3$. (Check $(0, 1)$: $2 \cdot 2 - 3 = 1$.)

Answer: (C)

18.

HARD · NO CALCULATOR NEEDED

The function g is defined by $g(x) = 3 \cdot 2^x$. For what value of a is $g(a + 2) = g(a) + 36$?

(A) 2 ✓

(B) 3

(C) 4

(D) 1

1. $g(a + 2) = 3 \cdot 2^{a+2} = 4 \cdot (3 \cdot 2^a) = 4g(a)$.
2. $4g(a) = g(a) + 36 \Rightarrow 3g(a) = 36 \Rightarrow g(a) = 12$.
3. $3 \cdot 2^a = 12 \Rightarrow 2^a = 4 \Rightarrow a = 2$.

Answer: (A)

19.

HARD · NO CALCULATOR NEEDED

The function f satisfies $f(3x - 2) = x^2 + 5$ for all values of x . What is the value of $f(10)$?

(A) 105

(B) 36

(C) 9

(D) 21 ✓

1. Find the input that makes $3x - 2 = 10$: $x = 4$.

2. $f(10) = f(3 \cdot 4 - 2) = 4^2 + 5 = 21$.

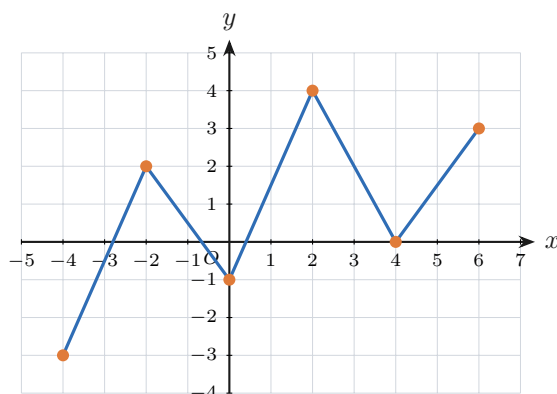
3. (Substituting 10 directly into $x^2 + 5$ would give 105, which is wrong because $x \neq 10$.)

Answer: (D)

20.

HARD · DESMOS

The graph of the piecewise linear function f is shown for $-4 \leq x \leq 6$. For how many values of x in this interval does $f(x) = 1$?



(A) 3

(B) 4

(C) 5 ✓

(D) 6

1. Draw the horizontal line $y = 1$.

2. It crosses the segment rising from $(-4, -3)$ to $(-2, 2)$, the segment falling to $(0, -1)$, the segment rising to $(2, 4)$, the segment falling to $(4, 0)$, and the segment rising to $(6, 3)$.

3. That is 5 crossings, so 5 values of x .

Answer: (C)

21.

HARD · NO CALCULATOR NEEDED

The table shows values of the function f . The function g is defined by $g(x) = 3f(x+1) - 2$. What is the value of $g(-1) + g(3)$?

x	0	2	4	6
$f(x)$	-1	3	7	2

(A) 6

(B) 14 ✓

(C) 18

(D) 22

$$1. g(-1) = 3f(0) - 2 = 3(-1) - 2 = -5.$$

$$2. g(3) = 3f(4) - 2 = 3(7) - 2 = 19.$$

$$3. g(-1) + g(3) = -5 + 19 = 14.$$

Answer: (B)**22.**

HARD · CALCULATOR

The function $V(t) = 24000(0.85)^t$ gives the value, in dollars, of a delivery van t years after it was bought. Which of the following statements must be true?

I. $V(t+1) = 0.85V(t)$ for every t .

II. $V(t+2) = 0.70V(t)$ for every t .

III. The van is worth less than \$12,000 when $t = 5$.

(A) II and III only

(B) I, II, and III

(C) I only

(D) I and III only ✓

$$1. \text{ I: } V(t+1) = 24000(0.85)^t(0.85) = 0.85V(t). \text{ True.}$$

$$2. \text{ II: } V(t+2) = 0.85^2V(t) = 0.7225V(t) \neq 0.70V(t). \text{ False (two 15\% drops are not a 30\% drop).}$$

$$3. \text{ III: } V(5) = 24000(0.85)^5 \approx 24000(0.4437) \approx 10,649 < 12,000. \text{ True.}$$

4. Statements I and III are true.

Answer: (D)

23.

HARD · NO CALCULATOR NEEDED

The function g is defined by $g(x) = \frac{1}{x^2 - 6x + k}$, where k is a constant. The function is undefined for exactly one value of x . What is the value of k ?

(A) 9 ✓

(B) 12

(C) 3

(D) 6

1. g is undefined where $x^2 - 6x + k = 0$.
2. Exactly one such x means the quadratic has one (repeated) root: discriminant = 0.
3. $(-6)^2 - 4(1)(k) = 0 \Rightarrow 36 = 4k \Rightarrow k = 9$. (Then $x^2 - 6x + 9 = (x - 3)^2$, zero only at $x = 3$.)

Answer: (A)**24.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The linear function f satisfies $f(f(x)) = 9x + 8$ for all x , and $f(1) = 5$. What is the value of $f(10)$?

1. Write $f(x) = ax + b$. Then $f(f(x)) = a(ax + b) + b = a^2x + (a + 1)b$.
2. Match $9x + 8$: $a^2 = 9$, so $a = 3$ or $a = -3$; and $(a + 1)b = 8$.
3. If $a = 3$: $4b = 8$, $b = 2$. If $a = -3$: $-2b = 8$, $b = -4$. Two candidates: $3x + 2$ and $-3x - 4$.
4. Use $f(1) = 5$: $3 + 2 = 5$ works, while $-3 - 4 = -7$ does not. So $f(x) = 3x + 2$.
5. $f(10) = 32$.

Answer: 32

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f satisfies $f(x) = f(8 - x)$ for all real x , and $g(x) = f(x + 4)$. If $g(3) = 11$ and $g(-1) = 5$, what is the value of $f(5) + f(7)$?

(A) 10

(B) 16 ✓

(C) 21

(D) 22

1. $g(3) = f(7) = 11$ and $g(-1) = f(3) = 5$.

2. By the symmetry $f(x) = f(8 - x)$: $f(5) = f(8 - 5) = f(3) = 5$.

3. $f(5) + f(7) = 5 + 11 = 16$.

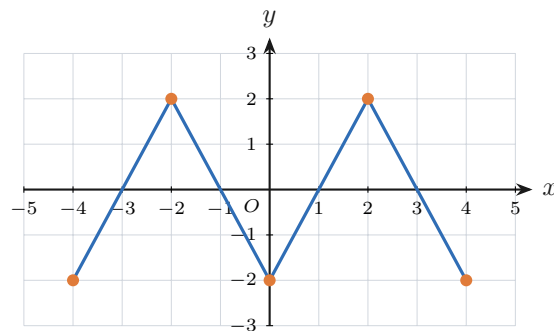
4. (Check with $f(x) = \frac{3}{4}(x - 4)^2 + \frac{17}{4}$: $f(3) = 5$, $f(5) = 5$, $f(7) = 11$.)

Answer: (B)

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

The graph of the piecewise linear function f is shown for $-4 \leq x \leq 4$. For how many values of x in this interval does $f(f(x)) = 0$?



1. First find where $f(u) = 0$: the graph crosses the x -axis at $u = -3, -1, 1, 3$.

2. So $f(f(x)) = 0$ requires $f(x) \in \{-3, -1, 1, 3\}$.

3. The outputs of f lie between -2 and 2 , so $f(x) = \pm 3$ is impossible.

4. $f(x) = -1$ is met once on each of the 4 segments: 4 solutions. $f(x) = 1$ likewise: 4 solutions.

5. Total: $4 + 4 = 8$.

Answer: 8

27.

SAT CHALLENGE · DESMOS

The function f is defined by $f(x) = -2|x - 1| + 8$. The graph of $y = f(x + 3) - 2$ and the line $y = k$ intersect at two points that are 9 units apart. What is the value of k ?

(A) 3

(B) 15

(C) -3 ✓

(D) -1

1. $f(x + 3) - 2 = -2|x + 2| + 8 - 2 = -2|x + 2| + 6$.

2. Set $-2|x + 2| + 6 = k \Rightarrow |x + 2| = \frac{6 - k}{2}$.

3. The two solutions are $-2 \pm \frac{6 - k}{2}$, which are $6 - k$ apart.

4. $6 - k = 9 \Rightarrow k = -3$.

5. (The horizontal shift does not change the distance; using the unshifted peak 8 would give -1.)

Answer: (C)

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

The functions f and g are defined only for $x = 1, 2, 3, 4, 5$, and their values are in the table. What is the sum of all values of x for which $f(g(x)) = g(f(x))$?

x	1	2	3	4	5
$f(x)$	3	5	1	2	4
$g(x)$	2	3	5	1	4

1. Compute both sides for each x (inner function first).

2. $x = 1$: $f(g(1)) = f(2) = 5$, $g(f(1)) = g(3) = 5$. Equal.

3. $x = 2$: $f(g(2)) = f(3) = 1$, $g(f(2)) = g(5) = 4$. Not equal.

4. $x = 3$: $f(g(3)) = f(5) = 4$, $g(f(3)) = g(1) = 2$. Not equal.

5. $x = 4$: $f(g(4)) = f(1) = 3$, $g(f(4)) = g(2) = 3$. Equal.

6. $x = 5$: $f(g(5)) = f(4) = 2$, $g(f(5)) = g(4) = 1$. Not equal.

7. The matching values are $x = 1$ and $x = 4$; their sum is 5.

Answer: 5

29.

SAT CHALLENGE · CALCULATOR

The function f is defined by $f(x) = ax^2 + 4x$, where a is a constant. The average rate of change of f over the interval $1 \leq x \leq 5$ is 22. What is the value of $f(3)$?

1. $f(1) = a + 4$ and $f(5) = 25a + 20$.
2. Average rate of change: $\frac{(25a + 20) - (a + 4)}{5 - 1} = \frac{24a + 16}{4} = 6a + 4$.
3. $6a + 4 = 22 \Rightarrow a = 3$.
4. $f(3) = 3(3)^2 + 4(3) = 27 + 12 = 39$.

Answer: 39**30.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = \frac{ax + 5}{x - 3}$ for $x \neq 3$, where a is a constant. If $f(f(x)) = x$ for every x in the domain of $f \circ f$, what is the value of $f(7)$?

(A) $\frac{13}{2}$ ✓

(B) $\frac{19}{4}$

(C) 26

(D) -4

1. Compute $f(f(x)) = \frac{a \cdot \frac{ax+5}{x-3} + 5}{\frac{ax+5}{x-3} - 3}$. Multiply top and bottom by $x - 3$: $\frac{a(ax + 5) + 5(x - 3)}{(ax + 5) - 3(x - 3)} = \frac{(a^2 + 5)x + 5a - 15}{(a - 3)x + 14}$.
2. Set this equal to x : $(a^2 + 5)x + 5a - 15 = (a - 3)x^2 + 14x$ for all x .
3. The x^2 term forces $a - 3 = 0$, so $a = 3$. (Then x -terms: $14x = 14x$; constants: $5(3) - 15 = 0$.)
4. $f(x) = \frac{3x + 5}{x - 3}$, so $f(7) = \frac{26}{4} = \frac{13}{2}$.

Answer: (A)

31.

SAT CHALLENGE · NO CALCULATOR NEEDED

For $x \geq 0$, let $f(x) = 120 - 90(0.8)^x$ and $g(x) = -3(x - 4)^2 + 125$. Which of the following statements is true?

(A) Neither attains a maximum

(B) Both f and g attain a maximum(C) Only f attains a maximum, 120**(D) Only g attains a maximum, 125 ✓**

1. g is a downward parabola with vertex $(4, 125)$ and $x = 4$ is in the domain, so g attains its maximum value 125.
2. $(0.8)^x > 0$ for every x , so $90(0.8)^x > 0$ and $f(x) < 120$ for every x . As x grows $f(x)$ gets closer to 120 but never equals it, so f has no maximum value even though it is bounded above by 120.
3. Only the third statement is correct.

Answer: (D)

32.

SAT CHALLENGE · DESMOS

$$2^x = 3x + k$$

In the equation above, k is a constant. What is the smallest integer value of k for which the equation has two distinct real solutions?

(A) 0

(B) -3

(C) -2 ✓

(D) -1

1. Rewrite as $g(x) = 2^x - 3x = k$. The equation has two solutions exactly when the horizontal line $y = k$ lies strictly above the minimum of g .
2. $g'(x) = 2^x \ln 2 - 3 = 0 \Rightarrow 2^x = \frac{3}{\ln 2} \approx 4.328$, so $x \approx 2.115$. Then $g_{\min} \approx 4.328 - 3(2.115) \approx -2.02$ (Desmos: minimum $\approx (2.115, -2.017)$).
3. Two solutions need $k > -2.017$; the smallest integer is -2 .
4. Check: for $k = -3$ the graphs of $y = 2^x$ and $y = 3x - 3$ never meet; for $k = -2$, $x = 2$ is one solution ($4 = 6 - 2$) and a second lies near $x \approx 2.2$.

Answer: (C)

33.

Let $g(x) = \log_2 x$ for $x > 0$ and $h(x) = \frac{x+8}{6}$. For which values of x is $h(x) > g(x)$?

(A) $x < 4$ or $x > 16$ **(B) $0 < x < 4$ or $x > 16$ ✓**(C) $4 < x < 16$ (D) $x > 16$ only

1. Look for intersections at powers of 2: $g(4) = 2 = h(4) = \frac{12}{6}$ and $g(16) = 4 = h(16) = \frac{24}{6}$.

2. The curve $y = \log_2 x$ is concave down, so a line can meet it in at most two points; these are the only two. They split $x > 0$ into three regions.

3. Test one value in each region. $x = 1$: $h = 1.5 > g = 0$. $x = 8$: $h = \frac{16}{6} \approx 2.67 < g = 3$. $x = 32$: $h \approx 6.67 > g = 5$.

4. So $h(x) > g(x)$ for $0 < x < 4$ and for $x > 16$.

Answer: (B)**34.**

The table gives values of a quadratic function g at equally spaced inputs.

x	2	5	8	11
$g(x)$	44	95	200	359

What is the value of $g(17)$?

(A) 677

(B) 572

(C) 785

(D) 839 ✓

1. Inputs step by 3. First differences of g : $95 - 44 = 51$, $200 - 95 = 105$, $359 - 200 = 159$.

2. Second differences: $105 - 51 = 54$ and $159 - 105 = 54$, constant, as expected for a quadratic.

3. Next first difference: $159 + 54 = 213$, so $g(14) = 359 + 213 = 572$.

4. The one after that: $213 + 54 = 267$, so $g(17) = 572 + 267 = 839$.

5. (Check: $g(x) = 3x^2 - 4x + 40$ gives $g(17) = 867 - 68 + 40 = 839$. Option 572 is $g(14)$; 677 keeps the last difference 159 constant; 785 adds 213 twice.)

Answer: (D)